

Suggested FYP topics for 2026-27

by

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Project code :	HFY1
Project title :	Network Coding over Multipath Networks
No. of students :	1 group (of 1-2 students) project
Description of project :	In traditional computer networks, although there might be multiple paths connecting the source and the destination node, only one path is used for data transmission. It is possible to increase the throughput by utilizing non-overlapped paths simultaneously. The separation and merge of data streams can be managed by network coding so that the focus is to discover available paths, control individual transmission rates, and design protocols to achieve this goal. Students are expected to: 1. Design/use protocols for multipath routing. 2. Design a scheme to utilize multiple paths. 3. Implement simple network coding schemes. Apply network simulator / conduct experiments to collect performance data
Industrial collaboration :	No.

Project code :	HFY2
Project title :	Optimizations in Batched Network Coding
No. of students :	1 group (of 1-2 students) project
Description of project :	Batched network coding is a branch of network coding that only requires small memory and computational resources at the intermediate nodes. There are still many open problems about decision making and optimizations at the source and the intermediate network nodes. Students may investigate one or more open problems. They can choose their preferred techniques according to their abilities, such as analyzing mathematical models directly, making decisions by machine learning, or designing and verifying schemes via simulations.
Industrial collaboration :	No.

Project code :	HFY3
Project title :	Experiments in Steganography and Information Hiding
No. of students :	1 group (of 1-2 students) project
Description of project :	Steganography is an art of hiding secret messages in other media such as images and audios. Students may investigate various

	topics in steganography, including implementing and/or designing steganographic schemes, performing steganalysis via machine learning and/or statistics, etc. Students are expected to have a mild background or self-learning the foundation of error-correcting codes and statistical analysis.
Industrial collaboration :	No.

Project code :	HFY4
Project title :	Self-Proposed Topics
No. of students :	1 group (of 1-2 students) project
Description of project :	Students may propose any topic, preferably on computer networking, Internet of Things, data compression, operating systems, network coding, communication, or optimizations. Students are expected to produce some novel results or schemes either in application or in theory.
Industrial collaboration :	No.

2026 March 9